

Chapter 3 Worksheet 3

Section 3.1: Measures of Center and Location

- a) **3-2** The following data reflects the number of defects produced on assembly line in the last 8 days (**Self-study**)

3	0	2	0	1	3	5	2
5	1	3	0	0	1	3	3
4	3	1	8	4	2	4	0

- b) Compute the mean number of defects
- c) Compute the median number of defects
- d) Determine if there is a mode, if so then indicate its value
- e) Compute the Q1, Q2 and Q3

3.6 Examine the following data

23	65	45	19	35	28	39	100	50	26	25	27
24	17	12	106	23	19	39	70	20	18	44	31

- a) Compute the Quartiles
- b) Calculate the 90th percentile
- c) Write down the **five-number summary** and **draw the Box and Whisker plot**
- d) Calculate 20th and 30th percentile
- e) are there any outlier?

3.13 The following data are selection of 30 Toys R Us outlet, sorted

18	19	21	22	24	25	25	26	27	28
29	29	30	32	33	33	35	35	36	37
37	37	37	37	38	40	41	41	44	46

- a) Compute Mean, Median and mode
- b) Indicate the shape of the distribution
- c) Construct Box and whisker plot for the sample (**Self-study**)

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3-4 The ages of 15 employees selected from the company are as follows: (Self-study)

53	32	61	27	39
44	45	57	25	28
33	35	36	33	41

- 1- Calculate Mean, Median, Q1 and Q3
- 2- What is the shape of the distribution is it symmetric or skewed

3.5 A random sample of the miles driven by 20 rental car customers is shown as follows (Self-study)

90	85	100	150
125	75	50	100
75	60	35	90
100	125	75	85
50	100	50	80

- 1- Compute the five-number summary.
- 2- **Draw the Box and whisker plot.**
- 3- Are there any outliers.

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Section 3.2: Measure of Variation

3.26 The following data shows the number of times a population of business flew on business during the previous month

4	6	9	4	5	7
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- a) Compute the range of these data
- b) Compute the variance and standard deviation S.D.
- c) Assuming these are sample, compute the variance and standard deviation, discuss the difference

3.30 Consider the following two samples (sorted from small to large)

Sample 1	10	12	15	20	25	27	27	31	35	37
Sample 2	1	2	3	4	16	16	16	16	18	118

- a) Compute the range, variance and standard deviations and interquartile range of each sample set
- b) Which data is more spread out based on these statistics

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3.31 The following set of data are the ages of people who attended five day seminar

45	21	38	48	28	34	26	53	29	27
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- a) Calculate the range and interquartile range for the data set
- b) Indicate whether the data set is from a population or sample. Calculate standard deviation
- c) Interpret your findings for part a and b

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Section 3.3: Using The Mean and Standard Deviation Together

3.48. The mean time that a certain model of light bulb will last is 400 hours, with a standard deviation equal to 50 hours.

- a. Calculate the standardized value for a light bulb that lasts 500 hours.
- b. Assuming that the distribution of hours that light bulbs last is **bell shaped**, answer the following questions:
 - I. what percentage of bulbs could be expected to last between 300 hours and 500 hours?
 - II. what percentage of bulbs could be expected to last longer than 500 hours?
 - III. what percentage of bulbs could be expected to last between 350 hours and 450 hours?

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3.50 You are given the following parameter for two populations

Population 1	Population 2
$\mu = 700$	$\mu = 29,000$
$\sigma = 50$	$\sigma = 5,000$

- Compute the coefficient of variation for each population
- Based on the answers in part a, which population has data values that are more variable relative to the size of the population mean?

3.53 If sample mean is 1,000 and the sample standard deviation is 250, determine the standardized value for the following:

- $x = 800$
- $x = 1,200$
- $x = 1,000$

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3.51 Two distribution of data are being analyzed. Distribution A has mean of 500 and standard deviation (S.D.) equal to 100. Distribution B has mean of 10 and standard deviation (S.D.) equal to 4.0. Based on this information, use coefficient of variation to determine which distribution has greater relative variation. (**Self-Study**)

3.3.7 The number of days that homes stay on the market before they sell in Seattle is bell-shaped with a mean equal to 8 days. Further, 95 percent of all homes are on the market between 2 and 14 days. Based on this information, what is the standard deviation for the number of days that houses stay on the market in Seattle?

3.3.8 A certain gym selected a random sample of 10 customers and monitored the number of times each customer used the workout facility in a one-month period. The data are shown in the table below. The gym's managers are considering a promotion in which they reward frequent users with a small gift. They have decided that they will only give gifts to those customers whose number of visits in a one-month period is 1 standard deviation above the mean. Find the minimum number of visits required to receive a gift.

16	18	18	19	13	20	18	12	20
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